

Plan

- o Decimals (R1.22)
- o Finite, countable, and uncountable sets. (R. first section of Ch 1)

How can we write down or store a real number?

Decimals

Let x be real.

Let n_0 be the largest integer s.t. $n_0 \leq X$.

Let n_1 be the largest integer s.t. $n_0 + \frac{n_1}{10} \leq X$

Let n_k be the largest integer s.t. $n_0 + \frac{n_1}{10} + \dots + \frac{n_k}{10^k} \leq X$

$n_0 . n_1 n_2 n_3 \dots$ is the decimal expansion of x .

$n_0 \in \mathbb{Z}$, $n_i \in \{0, \dots, 9\}$ for $i \geq 1$.

Prop Decimal expansion is a bijection from $\mathbb{R}$ to the
Set of sequences $n_0, n_1, n_2, \dots$ with $n_0 \in \mathbb{Z}$ and $n_k \in \{1, \dots, 9\}$
Such that the sequence isn't eventually all 9s.

Proof Idea Given such a sequence $n_0, n_1, \dots$ we get a real number

$$X_{(n_0, n_1, \dots)} = \sup_{n \in \mathbb{N}} \left(\sum_{k=0}^n \frac{n_k}{10^k} \right). \quad \text{We should show this is inverse to decimal expansion.}$$

Storing a real number requires infinite memory.

Def Let $f: A \rightarrow B$ be a map of sets.

- The image of $E \subset A$ is

$$f(E) = \{f(a) \mid a \in E\}.$$

- The preimage of $E \subset B$ is

$$f^{-1}(E) = \{a \mid f(a) \in E\}.$$

We write $f^{-1}(b)$ for $f^{-1}(\{b\})$ for $b \in B$.

Def Let A and B be sets

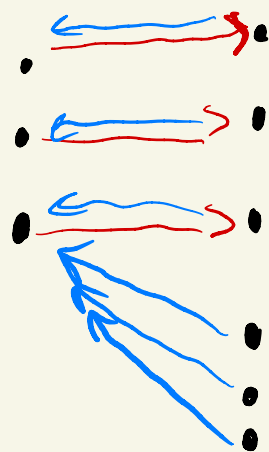
- $A \sim B$ if there exists a bijection $A \xrightarrow{\sim} B$. A, B have same cardinality.
- $A \leq B$ if there exists an injection $A \hookrightarrow B$. A has smaller cardinality than B .

Lemma For any two sets A, B

$\exists$ injection $A \hookrightarrow B \Leftrightarrow \exists$ surjection $B \twoheadrightarrow A$

Proof sketch

From an injection,
build a surjection the other way



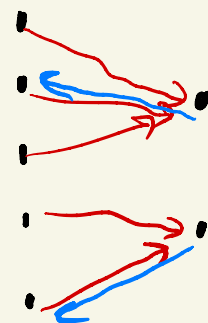
Let $f: A \hookrightarrow B$ be injective

Choose arbitrary $a_0 \in A$

Let $g(b) = \begin{cases} a \text{ st. } f(a) = b & \text{if } b \in f(A) \\ a_0 & \text{if } b \notin f(A) \end{cases}$

g is surjective.

From a surjection
build an injection the other way



Let $f: A \twoheadrightarrow B$ be surjective

Choose $a_b \in f^{-1}(b)$ for each $b \in B$
(uses axiom of choice)

Let $g(b) = a_b$ for all $b \in B$.

g is injective.

Theorem (Cantor-Schröder-Bernstein)

$$\bullet A \leq B \text{ and } B \leq A \Rightarrow A \sim B$$

Try to prove it! (not so easy)

This shows bijection classes of sets are an ordered set.

Def Let $\underline{n} = \{1, \dots, n\}$.

Def A is finite if $A \sim \underline{n}$ for some $n \in \mathbb{N}$.

We say $|A| = n$, and A has cardinality n .

Example If $|A| = m$, $|B| = n$ then

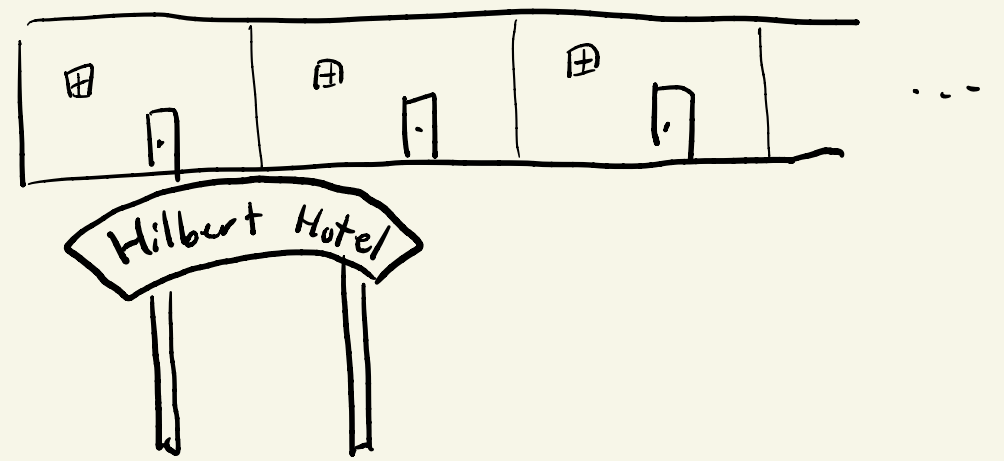
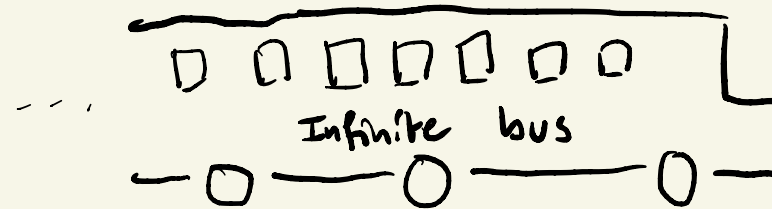
- $|A \times B| = mn$ product set
- $|A \cup B| \leq m + n$ union
- $|P(A)| = 2^m$ power set $P(A) = \{\text{subsets of } A\}$

Example $\mathbb{Z} \sim \mathbb{N}$

$$\mathbb{Z} = \{0, 1, -1, 2, -2, 3, -3, \dots\}$$

$$\phi: \mathbb{N} \rightarrow \mathbb{Z}$$

$$\phi(n) = \begin{cases} -\frac{n-1}{2} & n \text{ odd} \\ \frac{n}{2} & n \text{ even} \end{cases}$$



Def A is countable (countably infinite) iff $A \sim \mathbb{N}$.

Def A sequence of elements in a set A is a function $\phi: \mathbb{N} \rightarrow A$.

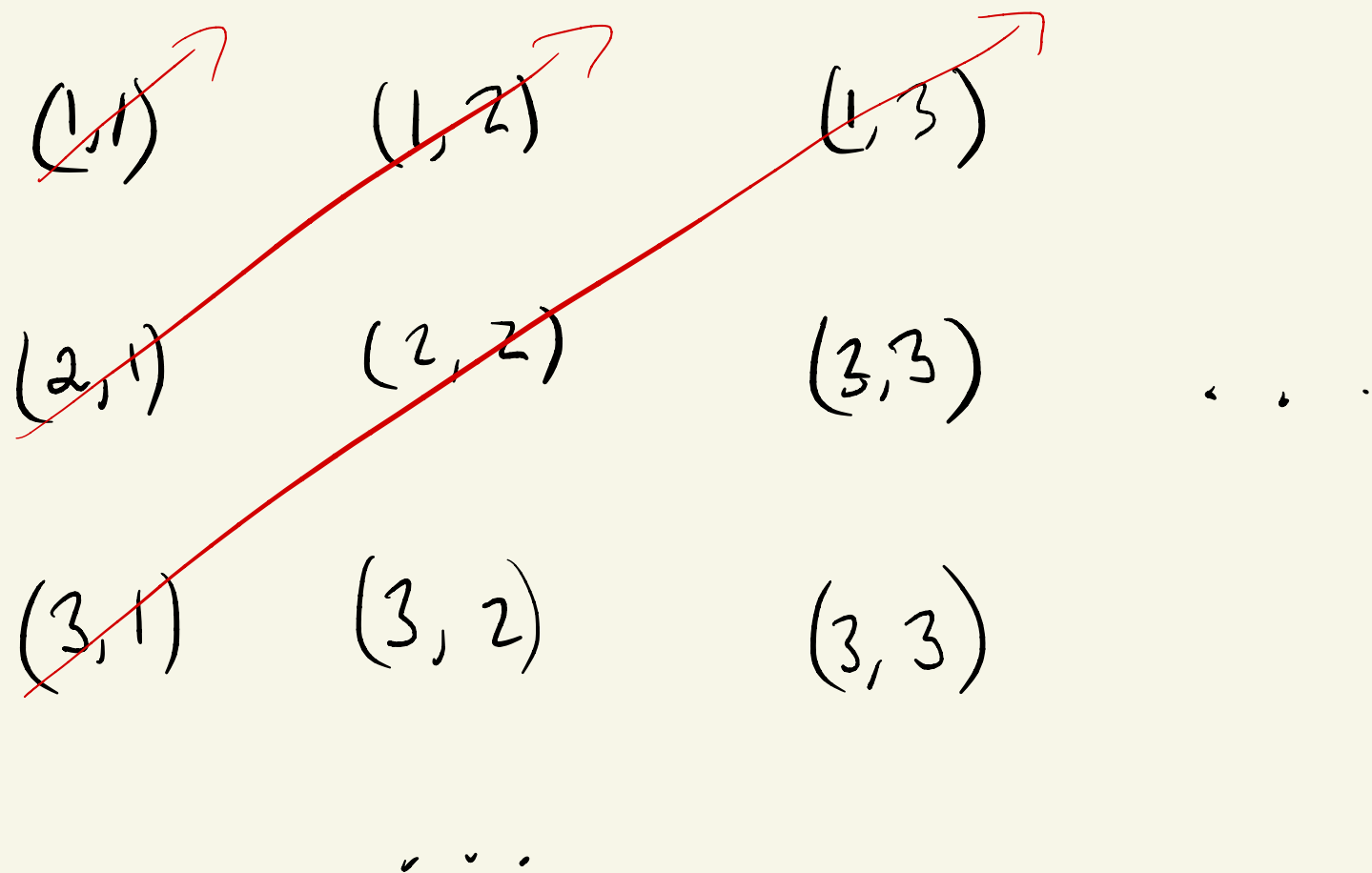
We write $x_n = \phi(n)$, and call the sequence $x_1, x_2, x_3, \dots$ or (x_n) .

Slight difference from Rudin's notation $\{x_n\}$.

"Arrange A into a sequence" means choose a bijection $\mathbb{N} \rightarrow A$.

Theorem $\mathbb{N}^2$ is countable.

proof



We can arrange $\mathbb{N}^2$ into a sequence.

$(1,1), (2,1), (1,2), (3,1), (2,2), (1,3), \dots$

Theorem A subset of a countable set is at most countable.

Proof: Let $E \subset A$. Suppose E is infinite.

Arrange elements of A into a sequence $\{x_n\}$.

$x_1, x_2, x_3, x_4, x_5, \dots$

$n_1=1 \quad n_2=3 \quad n_3=4 \quad \dots$

Let $n_1 \in \mathbb{N}$ be the smallest natural number s.t. $x_{n_1} \in E$

Having chosen $n_1, \dots, n_k$, let n_{k+1} be the next smallest natural number s.t. $x_{n_{k+1}} \in E$. We have arranged E into a sequence $\{x_{n_k}\}$.

Interpretation: no cardinality between finite and countable.

Theorem If $f: A \rightarrow B$ is surjective, and A is countable,
then B is at most countable.

Proof: We have shown there exists $B \rightarrow A$ injective,
so B is in bijection with a subset of A , thus is countable.

Theorem Let $\{E_n\}$ be a sequence of countable subsets of a set A . Then $\bigcup_{n \in \mathbb{N}} E_n = \{x \mid x \in E_n \text{ for some } n\}$ is countable.

proof: Arrange each E_n into a sequence $\{x_{nk}\}$.

$\phi: \mathbb{Z}^2 \longrightarrow \bigcup_{n \in \mathbb{N}} E_n$ is surjective, and $\mathbb{Z}^2$ is countable,

so $\bigcup_{n \in \mathbb{N}} E_n$ is countable

...

Theorem $\mathbb{Q}$ is countable.

Proof Let $E_n = \left\{ \frac{m}{n} \mid m \in \mathbb{Z} \right\}$

$$\mathbb{Q} = \bigcup_{n=1}^{\infty} E_n$$

$\mathbb{Q}$ is a countable union of countable sets.

Countable Sets

$\mathbb{N}$

$\mathbb{N} \times \mathbb{N}$

$\mathbb{Z}$

$\mathbb{N} \times \mathbb{N} \times \mathbb{N}$

$\mathbb{Q}$

What about $\mathcal{P}(\mathbb{N})$?

Thm Let A be the set of binary sequences $\mathbb{N} \rightarrow \{0, 1\}$.

A is uncountable.

Proof

Suppose $s_1, s_2, \dots$ was an arrangement of A into a sequence.

s_1	1	0	1	0	...
s_2	0	1	1	1	...
s_3	0	0	0	1	...
$\vdots$					

$t = 0, 0, 1, \dots$

Let $t \in A$ be the seq $1-s_{11}, 1-s_{12}, 1-s_{13}, \dots$.

$t \neq s_i$ for all $i \in \mathbb{N}$, so the sequence $\{s_n\}_{n \in \mathbb{N}}$ does not enumerate A . $\square$

Denote A by $\{0,1\}^{\mathbb{N}}$.

Corrolary, $\mathbb{R}$ is uncountable

proof: binary expansion:

$$\{t \mid 0 \leq t < 1\} \sim A \setminus \{\text{sequences ending in all 1's}\}$$

$$\{\text{sequences ending in all 1's}\} \sim \{0,1\} \cup \{0,1\}^2 \cup \{0,1\}^3 \cup \dots$$

is countable.

Cardinalities

$$\emptyset < \underline{1} < \underline{2} < \dots < \mathbb{N} < \{0,1\}^{\mathbb{N}}$$

are there more cardinalities?

Cantor's Continuum Hypothesis / First of Hilbert's 23 problems:

Is there a cardinality between $\mathbb{N}$ and $\{0,1\}^{\mathbb{N}}$?

Undecidable.

There are bigger ones though:

Thm There is no bijection $A \rightarrow P(A)$ for any non-empty A .

Proof Suppose $\phi: A \rightarrow P(A)$ was a bijection.

Let $E \subset A$ be the subset $\{x \in A \mid x \notin \phi(x)\}$.

Suppose $E = \phi(x)$ for some x .

Suppose $x \in E$. Then, $x \notin \phi(x) = E$. Contradiction

Suppose $x \notin E$. Then $x \notin \phi(x)$ so $x \in E$. contradiction.

So x is neither in or not in E . Contradiction.